A near-linear Chvátal–Erdős condition for Hamilton cycles in digraphs

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Abstract

For a digraph D, let $\alpha_2(D)$ be the largest size of a vertex set containing no directed 2-cycle. Let $f_2(a)$ be the least positive integer k such that every k-strongly connected digraph D with $\alpha_2(D) \le a$ has a Hamilton cycle. Jackson and Ordaz conjectured that $f_2(a) \le a+1$. Towards this conjecture, we establish the near-linear bound $f_2(a) = O\Big(\frac{a(\log a)^4}{(\log\log a)^2}\Big)$, and hence $f_2(a) = O_\varepsilon(a^{1+\varepsilon})$ for every fixed $\varepsilon > 0$. We also disprove the pancyclicity conjecture of Jackson and Ordaz that every digraph D with $\kappa(D) \ge \alpha_2(D)+1$ contains a directed cycle of every length from 2 to $|V(D)|$.

1 Introduction

All graphs are finite and simple, and all digraphs are finite, without loops or parallel arcs. Throughout this paper, we use G for a graph and D for a digraph. A graph G of order $n \ge 3$ is *Hamiltonian* if it contains a cycle through all its vertices, and *pancyclic* if it contains cycles of every length from 3 to n. The notion of pancyclicity was introduced by Bondy in [4]. In 1973, he proposed his famous meta-conjecture [5], which asserts that any non-trivial condition implying Hamiltonicity should also imply pancyclicity, up to a small class of exceptional graphs.

One such condition is the classical theorem of Chvátal and Erdős [9]. For a graph G, let $\alpha(G)$ denote its independence number, the maximum size of a set of pairwise nonadjacent vertices, and let $\kappa(G)$ denote its vertex-connectivity. The theorem states that every graph G of order at least three with $\kappa(G) \ge \alpha(G)$ is Hamiltonian. In the spirit of Bondy's meta-conjecture, Jackson and Ordaz [19] conjectured that every graph G with $\kappa(G) > \alpha(G)$ is pancyclic. The strict inequality is necessary, since balanced complete bipartite graphs satisfy $\kappa(G) = \alpha(G)$ and have no odd cycles.

Keevash and Sudakov [22] proved that $\kappa(G) \ge 600\alpha(G)$ implies pancyclicity, establishing the conjecture up to a constant factor. Draganić, Munhá Correia and Sudakov [10] subsequently obtained an asymptotically optimal result, showing that $\kappa(G) \ge (1+o(1))\alpha(G)$ already suffices for pancyclicity. More recently, Letzter [26] proved the conjecture for all sufficiently large graphs, retaining the condition $\kappa(G) > \alpha(G)$ without an asymptotic loss. A closely related and important problem was posed by Erdős [13] in 1972. He asked how large a Hamiltonian graph must be, in terms of its independence number, to guarantee pancyclicity, and conjectured that order $\Omega(\alpha(G)^2)$ should suffice. Draganić, Munhá Correia and Sudakov [11] resolved this conjecture by proving that every Hamiltonian graph with independence number at most k and at least $(2+o(1))k^2$ vertices is pancyclic, and this bound is asymptotically best possible.

Hamiltonicity and cycles of different lengths have also been extensively studied in digraphs; see the surveys [19, 25]. A digraph is *Hamiltonian* if it contains a directed cycle through every vertex, and *pancyclic* if it contains a directed cycle of every length from 2 to its order. A

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nonempty digraph is *strongly connected* if it contains a directed path from u to v for every ordered pair of vertices u, v, allowing a path of length zero when $u = v$. Ghouila-Houri [15] proved that a strongly connected n-vertex digraph is Hamiltonian if the sum of the indegree and outdegree of each vertex is at least n. Häggkvist and Thomassen [16] showed that the same degree condition also guarantees pancyclicity, with the complete balanced bipartite digraph having both arcs between every pair of vertices in different parts as the only possible exception.

Sharp degree conditions have also been obtained for oriented graphs, that is, digraphs with no pair of oppositely directed arcs. Keevash, Kühn and Osthus [21] proved that every sufficiently large n-vertex oriented graph with minimum indegree and outdegree at least $(3n - 4)/8$ is Hamiltonian, and that this bound is best possible. More recently, Lo, Patel and Yıldız [27] proved that every sufficiently large d-regular oriented graph on $n \leq 4d+1$ vertices is Hamiltonian, where d-regular means that each vertex has indegree and outdegree d.

The Chvátal–Erdős theorem suggests seeking analogous conditions in terms of connectivity and independence in digraphs. For a positive integer k, a digraph D is *k-strongly connected* if $|V(D)| \geq k + 1$ and $D - S$ is strongly connected for every $S \subseteq V(D)$ with $|S| < k$. We write $\kappa(D)$ for the largest such k, and set $\kappa(D) = 0$ if D is not strongly connected or has at most one vertex. For a digraph D, its *2-cycle graph* $G_2(D)$ is the graph on $V(D)$ in which two vertices are adjacent precisely when they form a directed 2-cycle in D. We define $\alpha_2(D) = \alpha(G_2(D))$; equivalently, $\alpha_2(D)$ is the largest size of a vertex set containing no directed 2-cycle. Replacing every edge of a graph by two opposite arcs preserves its connectivity and turns its independence number into $\alpha_2(D)$. Thus α_2 gives a natural directed analogue of the independence number in Chvátal–Erdős type conditions; see [3, 19, 25].

For a positive integer a, let $f_2(a)$ be the least positive integer k such that every k-strongly connected digraph D with $\alpha_2(D) \leq a$ is Hamiltonian. Jackson and Ordaz [19] conjectured that $\kappa(D) \geq \alpha_2(D) + 1$ guarantees a Hamilton cycle, which would imply $f_2(a) \leq a + 1$. Jackson [18] proved that f_2 exists and obtained $f_2(a) \leq 2^a(a + 2)!$. Kühn and Osthus [25] asked whether f_2 admits a polynomial upper bound. Ai and Shi [1] recently proved that $f_2(a) \leq 2a^3 + 2$. On the other hand, replacing every edge of $K_{a-1,a}$ by two opposite arcs gives a digraph D with $\kappa(D) = a - 1$ and $\alpha_2(D) = a$ that is not Hamiltonian, since every directed cycle meets the two parts in the same number of vertices. Hence $f_2(a) \geq a$ for $a \geq 2$. We prove a near-linear upper bound, within a polylogarithmic factor of the bound conjectured by Jackson and Ordaz.

Theorem 1.1. *There is an absolute constant $c > 0$ such that, for every integer $a \geq 3$,*

$$f_2(a) \leq c\frac{a(\log a)^4}{(\log\log a)^2}.$$

All unsubscripted logarithms are natural. We first prove a more general bound depending explicitly on the length of the even cycles used in the proof. For integers $a \geq 2$ and $k \geq 2$, let $\ell = \lceil \log_2 a \rceil + 1$ and

$$r_k(a) = \left\lceil (2(k-1)a)^{\frac{k}{k-1}} + \left(32k + \frac{k}{k-1}\right) a \right\rceil, \qquad h_k(a) = 2^{18}k^2a\ell^2.$$

Theorem 1.2. *For all integers $a \geq 2$ and $k \geq 2$ with $k \not\equiv 0 \pmod 3$, we have $f_2(a) \leq r_k(a) + h_k(a)$.*

From Theorem 1.1, we obtain the following corollary.

Corollary 1.3. *For every fixed $\varepsilon > 0$, we have $f_2(a) = O_\varepsilon(a^{1+\varepsilon})$.*

Jackson and Ordaz [19] also conjectured that every digraph D with $\kappa(D) \geq \alpha_2(D) + 1$ is pancyclic. We disprove this conjecture. In fact, even a linear connectivity condition in terms of $\alpha_2(D)$ does not guarantee pancyclicity.

Theorem 1.4. *There is no constant $c > 0$ such that every Hamiltonian digraph D satisfying $\kappa(D) \geq c\alpha_2(D)$ is pancyclic.*

Proof sketch of Theorem 1.1. We first prove Theorem 1.2. Let D satisfy $\alpha_2(D) \leq a$ and $\kappa(D) \geq r_k(a) + h_k(a)$, and let $G = G_2(D)$. Choose a maximal family $\mathcal{C}$ of pairwise vertex-disjoint copies of C_{2k} in G. Let $R = V(G) \setminus \bigcup_{C \in \mathcal{C}} V(C)$ be the remaining vertex set. Then $G[R]$ is C_{2k}-free. Motivated by the work of Chen, Faudree and Gould [8], we apply Pikhurko's even-cycle Turán bound and the Caro–Wei inequality to obtain $|R| \leq r_k(a)$. Hence $D - R$ is $h_k(a)$-strongly connected.

For each $C \in \mathcal{C}$, we choose a retained edge e_C and let P_C be the complementary path on the remaining $2k - 2$ vertices. The main ingredient is Lemma 3.1, which allows us to choose these edges, together with a direction and a colour for each P_C, so that the corresponding auxiliary digraph D_j is a-strongly connected for every colour j and has the required degree properties. A directed path in D_j expands to a directed path in D by concatenating whole complementary paths of colour j. The proof uses Chernoff bounds and the Caro–Wei inequality for the degree estimates, and combines strong connectivity with negative association to rule out directed separations of order less than a in D_j.

The vertices of R, together with the retained edges, can be covered by at most a pairwise vertex-disjoint directed paths. We now use one new colour in each round. By Lemma 4.2, we join the current paths in pairs, reducing their number from p to $\lceil p/2 \rceil$. After at most $\lceil \log_2 a \rceil$ rounds, one path remains, and we use a final colour to close it into a directed cycle C^*. This cycle covers R and every retained edge, and uses each complementary path either in its entirety or not at all. For each complementary path P_C still outside C^*, we replace the retained edge e_C by the other directed path around C between its endpoints. These replacements give a Hamilton cycle and prove Theorem 1.2. Finally, Theorem 1.1 follows from Theorem 1.2 by optimizing the choice of k.

Organization. In Section 2, we introduce the notation and collect the preliminary results. In Section 3, we prove the path-selection lemma. In Section 4, we prove Theorem 1.2 and derive Theorem 1.1. In Section 5, we construct the counterexamples to the Jackson–Ordaz pancyclicity conjecture.

Declaration of AI Use. The authors are grateful to AI tools for valuable assistance in exploring some of the proof strategies and constructing examples. This project began in July 2026. Early discussions with ChatGPT 5.6 explored several approaches, including linear expanders and the regularity lemma, but yielded only partial results. The authors compiled their reading notes and records of earlier attempts and shared them with the AI tools as background for subsequent discussions. The main progress came on 9 September 2026, when extensive author-guided discussions with ChatGPT 6 led to the near-linear bound stated in Theorem 1.1. Starting from an initial ten-page draft, the authors completely rewrote the Introduction and reorganized and partly expanded the proofs, producing the present seventeen-page manuscript. The authors directed the research process, independently verified all mathematical arguments and calculations, and take full responsibility for the mathematical correctness and exposition of the paper.

2 Preliminaries

We follow the graph and digraph terminology of [3, 6]. We write $V(G)$ and $E(G)$ for the vertex and edge sets of a graph, and $V(D)$ and $A(D)$ for the vertex and arc sets of a digraph. The order of a graph or digraph is its number of vertices, and its size is its number of edges or arcs. A directed path visits distinct vertices in an order consistent with its arcs, a directed cycle does the same cyclically and has length at least two. For $U \subseteq V(D)$, the induced subdigraph $D[U]$ keeps exactly the vertices in U and all arcs of D with both endpoints in U, and $D - U = D[V(D) \setminus U]$;

induced subgraphs of an undirected graph are defined in the same way. For $v \in V(D)$, the outneighbourhood and inneighbourhood are

$$N_D^+(v) = \{w \in V(D) : vw \in A(D)\}, \qquad N_D^-(v) = \{w \in V(D) : wv \in A(D)\}.$$

Let $d_D^+(v) = |N_D^+(v)|$ for $v \in V(D)$, and, when D is nonempty, let $\delta^+(D) = \min_{v \in V(D)} d_D^+(v)$. For $v \in V(D)$ and $U \subseteq V(D)$, let $d_U^+(v) = |N_D^+(v) \cap U|$. The corresponding notation $d_D^-(v)$, $\delta^-(D)$, and $d_U^-(v)$ is defined using inneighbours. The notation $d_U^+(v)$ and $d_U^-(v)$ does not require $v \in U$. The tail and head of an arc uv are u and v, respectively. A source is a vertex of indegree zero, and a sink is a vertex of outdegree zero. For an undirected graph G, let $N_G(v) = \{w : vw \in E(G)\}$, $N_G[v] = N_G(v) \cup \{v\}$ and $d_G(v) = |N_G(v)|$.

The length of a path or cycle is its number of edges or arcs, and the internal vertices of a path are all its vertices other than its endpoints. A *path cover* of a graph or digraph H is a family of pairwise vertex-disjoint paths whose vertex sets partition $V(H)$; for a digraph, the paths are directed. A *linear forest* is a graph whose components are paths, including isolated vertices, and it is *spanning* in G if it has vertex set $V(G)$. The endpoints of a linear forest are the vertices of degree at most one, and every path in $G_2(D)$ can be directed in either order in D. For a graph H, we write $\mathrm{ex}(n, H)$ for the largest number of edges in an n-vertex graph containing no subgraph isomorphic to H, not necessarily an induced subgraph. We use $[s] = \{1, \dots, s\}$ and $A \dot\cup B$ for a disjoint union.

If D is k-strongly connected, then every vertex has at least k inneighbours and k outneighbours. Indeed, if a vertex had fewer than k outneighbours, deleting them would leave it unable to reach another remaining vertex, which exists because $|V(D)| \geq k+1$. The inneighbour assertion is analogous. Also, deleting a set S with $|S| = s < k$ leaves a $(k-s)$-strongly connected digraph, because any further deletion of fewer than $k-s$ vertices removes fewer than k vertices in total, and at least $k-s+1$ vertices remain before that further deletion.

Theorem 2.1 (Caro–Wei [7, 29])**.** *Every nonempty graph G of order n and size m satisfies*

$$\sum_{v \in V(G)} \frac{1}{d_G(v)+1} \leq \alpha(G), \qquad \frac{n^2}{2m+n} \leq \alpha(G).$$

Theorem 2.2 (Pikhurko [28])**.** *For all integers $k \geq 2$ and $n \geq 1$, we have*

$$\mathrm{ex}(n, C_{2k}) \leq (k-1)n^{1+\frac{1}{k}} + 16(k-1)n.$$

Lemma 2.3. *Let $a \geq 2$ and $k \geq 2$ be integers. If a graph G contains no C_{2k} and satisfies $\alpha(G) \leq a$, then $|V(G)| \leq r_k(a)$.*

Proof. Let $n = |V(G)|$. If $n = 0$, the conclusion is immediate. Hence assume that $n \geq 1$, and let $m = |E(G)|$. By Theorem 2.1, $n^2 \leq a(2m+n)$, while Theorem 2.2 gives $2m \leq 2(k-1)n^{1+\frac{1}{k}} + 32(k-1)n$. Substituting the second bound into the first and dividing by n yields

$$n \leq 2(k-1)an^{\frac{1}{k}} + (32(k-1)+1)a. \tag{2.1}$$

Let $x_0 = (2(k-1)a)^{\frac{k}{k-1}}$. Then $2(k-1)a\, x_0^{\frac{1}{k}} = x_0$. If $n \leq x_0$, the conclusion follows. Otherwise, let $t = \frac{n-x_0}{x_0} > 0$, so $n = x_0(1+t)$. The binomial theorem gives $(1+\frac{t}{k})^k \geq 1+t$, and hence $(1+t)^{\frac{1}{k}} \leq 1+\frac{t}{k}$. Consequently, $2(k-1)an^{\frac{1}{k}} \leq x_0(1+\frac{t}{k}) = x_0 + \frac{n-x_0}{k}$. Using this in (2.1), we obtain $\frac{k-1}{k}(n-x_0) \leq (32(k-1)+1)a$ and so

$$n \leq x_0 + \left(32k + \frac{k}{k-1}\right) a \leq r_k(a).$$

This proves the lemma. □

Gallai and Milgram [14] proved that every digraph has a path cover of size at most the independence number of its underlying graph. For undirected graphs, the corresponding bound follows from the standard endpoint-joining argument: in a path cover with the minimum number of components, endpoints belonging to distinct paths are nonadjacent; see, for example, Han [17]. We shall use the following version, which keeps a prescribed spanning linear forest.

Lemma 2.4. *Let G be a graph and let F be a spanning linear forest of G. There is a spanning linear forest F' of G containing every edge of F, with at most $\alpha(G)$ components, such that every endpoint of F' is an endpoint of F and every component of F is a subpath of a component of F'.*

Proof. Starting from F, repeatedly join two distinct components whenever an endpoint of one is adjacent in G to an endpoint of the other. The union of the two paths and this joining edge is again a path, including when either path is a single vertex. Thus every operation preserves all existing edges, reduces the number of components by one, and creates no new endpoint. Since the graph is finite, the process stops at a spanning linear forest F' whose endpoints in different components are pairwise nonadjacent. Choosing one endpoint from each component gives an independent set, so there are at most $\alpha(G)$ components. □

A *matching* is a set of edges with pairwise disjoint endpoints, and a *vertex cover* is a set of vertices meeting every edge.

Theorem 2.5 (Kőnig's theorem; see [6])**.** *Let G be a bipartite graph. Then the maximum size of a matching in G equals the minimum size of a vertex cover of G.*

For disjoint vertex sets A, B of a digraph, a *matching of arcs from A to B* consists of arcs with tails in A, heads in B, and pairwise disjoint endpoints.

Lemma 2.6. *Let D be a k-strongly connected digraph, where $k \geq 1$, and let $V(D) = A \dot\cup B$ with $A, B \neq \emptyset$. Then D has a matching of arcs from A to B of size at least $\min\{k, |A|, |B|\}$.*

Proof. Form the bipartite graph H with parts A, B in which uv is an edge precisely when $u \in A$, $v \in B$, and $uv \in A(D)$. If its maximum matching had size $m < \min\{k, |A|, |B|\}$, Theorem 2.5 would give a vertex cover S of size m. Since $|S| < |A|, |B|$, both $A \setminus S$ and $B \setminus S$ are nonempty, and the cover property says that no arc goes from the former to the latter in $D - S$. Any directed path from $A \setminus S$ to $B \setminus S$ would have a first arc crossing this partition, so no such path exists. Thus $D - S$ is not strongly connected, contradicting $|S| < k$. □

We write $\mathbb{P}$ for probability and $\mathbb{E}$ for expectation, while A^c denotes the complementary event of A. We shall use Markov's inequality in the form $\mathbb{P}[X \geq t] \leq \frac{\mathbb{E}[X]}{t}$ for every nonnegative random variable X and every $t > 0$. A Bernoulli random variable takes values 0 and 1.

A finite random vector $(Z_i)_{i \in I}$ is *negatively associated* if, for disjoint subsets $I_1, I_2 \subseteq I$ and coordinatewise nondecreasing functions f, g,

$$\mathbb{E}\big[f((Z_i)_{i \in I_1}) g((Z_i)_{i \in I_2})\big] \leq \mathbb{E}\big[f((Z_i)_{i \in I_1})\big]\ \mathbb{E}\big[g((Z_i)_{i \in I_2})\big]\,,$$

whenever the expectations exist. Here coordinatewise nondecreasing means that increasing a coordinate while fixing the others cannot decrease the function value. The following three standard properties of negative association are due to Joag-Dev and Proschan [20]. See also Dubhashi and Ranjan [12].

Proposition 2.1 (Joag-Dev and Proschan [20])**.**

(i) *A finite 0–1 random vector with at most one coordinate equal to 1 is negatively associated.*

(ii) *The concatenation of finitely many mutually independent, negatively associated random vectors is negatively associated.*

(iii) *Let $(Z_i)_{i\in I}$ be negatively associated, let $I_1,\dots,I_t \subseteq I$ be pairwise disjoint, and let $f_1,\dots,f_t$ be coordinatewise nondecreasing functions. Then the random vector $\big(f_j((Z_i)_{i\in I_j})\big)_{j\in[t]}$ is negatively associated.*

If Z is a sum of independent or negatively associated Bernoulli random variables with mean $\mu = \mathbb{E}[Z]$, then the Chernoff bound gives $\mathbb{P}\big[Z \le \frac{\mu}{2}\big] \le e^{-\mu/8}$. See [2] for the independent case and [12] for the negatively associated case. The next lemma applies this bound together with Proposition 2.1.

Lemma 2.7. *Let $M = \{u_iv_i : i \in [r]\}$ be a matching from a set A to a disjoint set B, and partition A and B separately into parts. In each part select at most one vertex, independently of all other parts, and let p_i, q_i be the probabilities that u_i, v_i are selected, respectively. Call an edge $u_iv_i \in M$* surviving *if both u_i and v_i are selected. If Y is the number of surviving edges and $\mu = \sum_{i=1}^r p_iq_i$, then*

$$\mathbb{P}\left[Y \le \frac{\mu}{2}\right] \le e^{-\mu/8}.$$

Moreover, the surviving edges meet distinct parts on each side.

Proof. Let X_i indicate that u_i is selected and let Z_i indicate that v_i is selected. Within each part the full vector of vertex-selection indicators has at most one 1, so it is negatively associated by Proposition 2.1(i). Since the selections in different parts are independent, the full vector of all selection indicators is negatively associated by Proposition 2.1(ii).

For each i, let Y_i be the indicator that u_iv_i survives. Then $Y_i = X_iZ_i$. Because M is a matching, the coordinate pairs $\{X_i, Z_i\}$ are disjoint for distinct i, although different coordinates may correspond to vertices in the same part. Each product is nondecreasing in its two coordinates, so Proposition 2.1(iii) implies that $(Y_i)_{i\in[r]}$ is negatively associated.

Since u_i and v_i belong to distinct parts, their selections are independent. Thus $\mathbb{E}[Y_i] = p_iq_i$ and $\mathbb{E}[Y] = \sum_{i=1}^r p_iq_i = \mu$. The Chernoff bound therefore gives $\mathbb{P}\big[Y \le \frac{\mu}{2}\big] \le e^{-\mu/8}$. Finally, the surviving edges meet distinct parts on each side, since each part supplies at most one selected vertex. □

3 A path-selection lemma

Let $a \ge 2$ and $t \ge 4$ be integers with $t \not\equiv 0 \pmod 3$, let $1 \le \ell \le 2a$ be an integer, and let

$$h = 2^{16}t^2a\ell^2, \qquad d_0 = \frac{h}{4t\ell}.$$

Let D satisfy $\alpha_2(D) \le a$, choose $R \subseteq V(D)$, and suppose that $D' = D - R$ is h-strongly connected, and let $U = V(D')$. Let $T \subseteq R$ be fixed before any random choices, with $|T| \le 2a$ and $d_U^+(v), d_U^-(v) \ge h$ for every $v \in T$. Assume that a family $\mathcal{C}$ of vertex-disjoint t-cycles in $G_2(D)$ covers exactly $V(D')$.

For each $C \in \mathcal{C}$, choose an edge $e_C \in E(C)$, called the *retained edge*, and delete its two endpoints from the cycle C to obtain a path P_C on $t-2$ vertices, called its *complementary path*. Choose one of the two directions of P_C and a colour in $[\ell]$, with choices independent between different cycles. For each colour j, let $\mathcal{P}_j$ be its family of oriented paths, and let x_P and y_P be the first and last vertices of an oriented path P, respectively. Let D_j be the auxiliary digraph whose vertices are v_P, one for each $P \in \mathcal{P}_j$, with arcs defined by

$$v_Pv_{P'} \in A(D_j) \quad\Longleftrightarrow\quad P \ne P' \text{ and } y_Px_{P'} \in A(D).$$

A directed path of D_j expands to a directed path of D by traversing each represented path in full and, for each pair of consecutive represented paths P, P', using the arc $y_Px_{P'}$ of D.

For $v \in V(D') \cup T$, let $d_j^+(v)$ count the paths $P \in \mathcal{P}_j$ with $vx_P \in A(D)$, and let $d_j^-(v)$ count those with $y_P v \in A(D)$. When $v \in V(D')$, let C be the unique cycle in $\mathcal{C}$ containing v, and exclude the path arising from C from these counts. When $v \in T$, no such exclusion is made.

Lemma 3.1. *With the notation and assumptions above, the retained edges, directions, and colours can be chosen such that, for every $j \in [\ell]$,*

$$\kappa(D_j) \geq a, \qquad d_j^+(v), d_j^-(v) \geq a \quad \text{for all } v \in V(D') \cup T.$$

Proof. Let $G = G_2(D)$. Independently for each $C \in \mathcal{C}$, choose its retained edge, path direction, and colour uniformly from the $2t\ell$ possibilities, and denote this choice by ξ_C. For $\sigma \in \{+,-\}$, we use $d_U^\sigma(v)$ and $d_j^\sigma(v)$ for the corresponding $+$ or $-$ quantities defined above.

Claim 1. *With probability greater than $\frac{15}{16}$, every $d_j^\sigma(v)$, for $j \in [\ell]$, $v \in V(D') \cup T$, and $\sigma \in \{+,-\}$, is at least d_0.*

Proof. Fix $v \in V(D') \cup T$, $j \in [\ell]$, and $\sigma \in \{+,-\}$, and let $d = d_U^\sigma(v) \geq h$. A fixed vertex z of a cycle C is an endpoint of P_C for exactly two choices of the retained edge. For each resulting path, exactly one orientation makes z its first vertex and exactly one makes z its last vertex. Thus z is the relevant endpoint of a colour-j path with probability $\frac{1}{t\ell}$, where the relevant endpoint is the first vertex when $\sigma = +$ and the last vertex when $\sigma = -$. If C does not contain v and contains s outneighbours of v when $\sigma = +$, or s inneighbours when $\sigma = -$, its contribution to $d_j^\sigma(v)$ is a Bernoulli variable of mean $\frac{s}{t\ell}$. Omitting the cycle containing v, when necessary, loses at most t such neighbours. The remaining contributions are independent, with total mean $\mu \geq \frac{d-t}{t\ell} \geq \frac{d}{2t\ell}$, since $d \geq h = 2^{16}t^2a\ell^2 \geq 2t$. Since $\frac{d}{4t\ell} \leq \frac{\mu}{2}$, the Chernoff bound gives

$$\mathbb{P}\left[d_j^\sigma(v) < \frac{d}{4t\ell}\right] \leq e^{-\frac{d}{16t\ell}}. \tag{3.1}$$

We next sum these bounds over $V(D')$ using the Caro–Wei inequality. Let $\beta = 16t\ell$. Every neighbour of v in $G[V(D')]$ is both an inneighbour and an outneighbour of v in D', so Theorem 2.1 gives

$$\sum_{v \in V(D')} \frac{1}{d_U^\sigma(v)+1} \leq \sum_{v \in V(D')} \frac{1}{d_{G[V(D')]}(v)+1} \leq a.$$

Since $h = 2^{16}t^2a\ell^2 \geq 16t\ell = \beta$, the inequality $1 + x \leq e^x$ gives, for $d \geq h$,

$$d + 1 = (h+1)\left(1 + \frac{d-h}{h+1}\right) \leq (h+1)e^{\frac{d-h}{\beta}}.$$

Consequently,

$$\sum_{v \in V(D')} e^{-\frac{d_U^\sigma(v)}{\beta}} \leq (h+1)e^{-\frac{h}{\beta}} \sum_{v \in V(D')} \frac{1}{d_U^\sigma(v)+1} \leq a(h+1)e^{-\frac{h}{\beta}}.$$

Let E_0 be the event in the claim, and let $x = \frac{h}{\beta} = 2^{12}ta\ell \geq 2^{15}$. Since $d_U^\sigma(v) \geq h$, we have $d_0 \leq \frac{d_U^\sigma(v)}{4t\ell}$. For each fixed $j \in [\ell]$ and $\sigma \in \{+,-\}$, combining (3.1) with the above bound gives

$$\sum_{v \in V(D')} \mathbb{P}[d_j^\sigma(v) < d_0] \leq a(h+1)e^{-x}.$$

Each vertex of T contributes at most e^{-x}, so $|T| \leq 2a$ and a union bound over all colours and both signs yield

$$\mathbb{P}[E_0^c] \leq 2\ell\big(a(h+1) + |T|\big)e^{-x} \leq 2a\ell(h+3)e^{-x}.$$

Since $h \geq 3$ and $h = 16t\ell x$, we have

$$2a\ell(h+3) \leq 4a\ell h = 64at\ell^2 x \leq 64x^3,$$

where the last inequality uses $at\ell \leq x$ and $\ell \leq x$. Moreover, $x \geq 2^{15} > 64 \cdot 384$ implies $e^{x/2} \geq \frac{(x/2)^4}{4!} = \frac{x^4}{384} \geq 64x^3$. Therefore,

$$\mathbb{P}[E_0^c] \leq 64x^3 e^{-x} \leq e^{-x/2} < \frac{1}{16}. \tag{3.2}$$

This proves the claim. □

On E_0, every D_j is nonempty, since $D' \neq \emptyset$ and $d_0 > 0$. For $v_P \in V(D_j)$, its outdegree is $d_j^+(y_P)$ and its indegree is $d_j^-(x_P)$, since the omitted cycle is precisely the cycle supplying P. Since $d_0 = \frac{h}{4t\ell} = 2^{14}ta\ell > a$, we have $|V(D_j)| \geq d_0 + 1 > a$. It remains to rule out a partition $V(D_j) = L' \dot\cup X \dot\cup R'$ with $L', R' \neq \emptyset$, $|X| < a$, and no arc from L' to R'; call this a *bad separation*. Indeed, if $D_j - X$ is not strongly connected, the vertices reachable from a vertex that cannot reach all of $D_j - X$ form a possible set L', and the remaining vertices form R'.

Fix a colour j and an ordering of $\mathcal{C}$. We examine the choices ξ_C one cycle at a time, following the fixed ordering. Each choice is fixed in advance and is not changed when its cycle is examined. To describe a possible bad separation, we build disjoint sets I_L, I_R, initially empty, whose union is independent in G. The vertices in I_L will be last vertices y_P of paths P with $v_P \in L'$, and those in I_R will be first vertices x_P of paths P with $v_P \in R'$. For the current sets I_L, I_R, let

$$A_L = \{v \in V(G) : N_G[v] \cap I_L \neq \emptyset,\ N_G[v] \cap I_R = \emptyset\},$$
$$A_R = \{v \in V(G) : N_G[v] \cap I_R \neq \emptyset,\ N_G[v] \cap I_L = \emptyset\}.$$

The sets A_L, A_R are disjoint and are redefined whenever I_L or I_R changes. We examine only a cycle not previously examined and contained in neither A_L nor A_R. A cycle contained in one of these sets at an earlier stage may therefore become available for examination later, as the sets change. For the complementary path P of a cycle being examined, the choices that we need to record are those satisfying

$$P \in \mathcal{P}_j, \qquad x_P \notin A_R, \qquad y_P \notin A_L. \tag{3.3}$$

Claim 2. *If a cycle C is contained in neither A_L nor A_R, at least one of its $2t\ell$ choices satisfies* (3.3).

Proof. Let $C = v_0 v_1 \cdots v_{t-1} v_0$, with indices taken modulo t. The possible endpoint pairs of a complementary path are precisely $\{v_i, v_{i+3}\}$ for $i \in \mathbb{Z}_t$. Let H be the graph on $V(C)$ whose edges are these pairs. Since $t \not\equiv 0 \pmod 3$, we have $\gcd(3,t) = 1$, so H is connected.

Suppose that no complementary path can be oriented with its first vertex outside A_R and its last vertex outside A_L. For each edge $zw \in E(H)$, both orientations then fail these requirements. As $A_L \cap A_R = \emptyset$, this implies that $\{z,w\} \subseteq A_L$ or $\{z,w\} \subseteq A_R$. The connectedness of H now implies that $V(C)$ is contained in A_L or in A_R, a contradiction. Choose a suitable orientation and assign colour j. This gives (3.3). □

Let $r_0 = 2a - 1$ and let $N_0 = \frac{h}{16t} = 2^{12}ta\ell^2$. A word of length r_0 over $\{\mathrm{L}, \mathrm{R}, \mathrm{X}\}$ will specify how to proceed whenever (3.3) holds. Fix such a word, start with $I_L = I_R = \emptyset$ and a counter $\eta_X = 0$, and always examine the first available cycle in the fixed ordering that is contained in neither A_L nor A_R. If its path fails (3.3), make no change. Otherwise read the next symbol of the word: L adds y_P to I_L, R adds x_P to I_R, and X increases η_X by one. Discard the word if the indicated addition destroys independence, if $\eta_X = a$, or if a further symbol is needed after all r_0 symbols have been used. After each examination, apply the preceding rules. Unless the word is

discarded, repeat the procedure until either N_0 cycles have been examined or every unexamined cycle is contained in A_L or in A_R. The next cycle and every decision in this procedure depend only on the fixed word and the choices on cycles already examined.

A bad separation *agrees* with the procedure if, whenever a symbol is used for a path P, that symbol is L, R, or X according as $v_P \in L'$, $v_P \in R'$, or $v_P \in X$.

Claim 3. *Every bad separation agrees with the procedure for some word that is not discarded. There are 3^{2a-1} possible words.*

Proof. Fix a bad separation. Whenever (3.3) holds for a path P, choose the next symbol to be L, R, or X according as $v_P \in L'$, $v_P \in R'$, or $v_P \in X$.

Suppose first that $v_P \in L'$, and let $y = y_P$. Every vertex of I_R is the first vertex of an earlier path represented in R'. An edge of G joining y to such a vertex would give an arc from L' to R' in D_j, contrary to the definition of a bad separation. Thus y has no neighbour in I_R. Moreover, since the current cycle has not been examined before and the cycles of $\mathcal{C}$ are vertex-disjoint, we have $y \notin I_R$. Hence $N_G[y] \cap I_R = \emptyset$. Together with $y \notin A_L$, this implies $N_G[y] \cap I_L = \emptyset$. Thus adding y to I_L preserves independence.

Similarly, suppose that $v_P \in R'$, and let $x = x_P$. If x were adjacent in G to a vertex of I_L, the corresponding arc would go from L' to R' in D_j, again a contradiction. Since the current cycle has not been examined before, $x \notin I_L$, and hence $N_G[x] \cap I_L = \emptyset$. Together with $x \notin A_R$, this gives $N_G[x] \cap I_R = \emptyset$. Thus adding x to I_R also preserves independence.

Each symbol L or R therefore adds a new vertex to the independent set $I_L \cup I_R$, so there are at most a such symbols. Each symbol X corresponds to a different vertex $v_P \in X$, so there are at most $a-1$ such symbols. Hence at most $a + (a-1) = 2a - 1 = r_0$ symbols are used, and the word is never discarded. If fewer than r_0 symbols are used, fill the remaining positions with X. These additional symbols are never read and do not affect the procedure. Since each of the $r_0 = 2a - 1$ positions has three possible symbols, there are 3^{2a-1} possible words. □

Claim 4. *For a fixed word, the probability that N_0 cycles are examined without discarding the word is at most $e^{-\frac{h}{256t^2\ell}}$.*

Proof. The next cycle is determined only by the choices on the cycles already examined, so each of its $2t\ell$ possible choices still has conditional probability $\frac{1}{2t\ell}$. By Claim 2, the conditional probability that (3.3) holds is therefore at least $p = \frac{1}{2t\ell}$.

Let Z_i indicate that this condition holds at the ith examination. If the procedure ends before N_0 examinations, extend the sequence to length N_0 using independent Bernoulli variables of mean p, independent of all original choices. The conditional success probability is at least p at every position. Taking conditional expectations, the same bound holds when conditioning only on the earlier indicators: $\mathbb{P}[Z_i = 1 \mid Z_1, \dots, Z_{i-1}] \geq p$. Since Z_i takes values 0 and 1, we have

$$\mathbb{E}[2^{-Z_i} \mid Z_1, \dots, Z_{i-1}] = 1 - \frac{1}{2}\mathbb{P}[Z_i = 1 \mid Z_1, \dots, Z_{i-1}] \leq 1 - \frac{p}{2}.$$

Let $S = \sum_{i=1}^{N_0} Z_i$. Iterating conditional expectations gives $\mathbb{E}[2^{-S}] \leq (1 - \frac{p}{2})^{N_0}$. Applying Markov's inequality to 2^{-S} with threshold $2^{-pN_0/2}$, and using $1 - \frac{p}{2} \leq e^{-p/2}$ and $\log 2 < \frac{3}{4}$, we obtain

$$\mathbb{P}\left[S \leq \frac{pN_0}{2}\right] = \mathbb{P}\left[2^{-S} \geq 2^{-pN_0/2}\right] \leq 2^{pN_0/2}\mathbb{E}[2^{-S}] \leq 2^{pN_0/2}\left(1 - \frac{p}{2}\right)^{N_0} \leq e^{-pN_0/8}.$$

Now suppose that N_0 cycles are examined without discarding the word. Each occurrence of $Z_i = 1$ uses one symbol of the word, so $S \leq r_0$. Since $pN_0 = \frac{h}{32t^2\ell} = 2^{11}a\ell \geq 2(2a-1) = 2r_0$, this event implies $S \leq pN_0/2$. Consequently, its probability is at most $e^{-pN_0/8} = e^{-\frac{h}{256t^2\ell}}$. □

Claim 5. *Fix a word and condition on the exact choices read in a procedure that stops before N_0 examinations without discarding the word. For every such event of positive probability, the conditional probability that some bad separation agrees with the procedure is at most*

$$2he^{-\frac{h}{16t\ell}} + e^{-\frac{h}{64t^2\ell^2}}.$$

Proof. Let E_1 denote the specified event, including the ordered list of examined cycles and their choices, and let $\mathcal{C}_0$ be this set of cycles. Changing choices on other cycles does not change the procedure: at each stage the next cycle and the decision to stop are determined by choices already read. Thus E_1 fixes only $(\xi_C)_{C\in\mathcal{C}_0}$, and the remaining choices are still independent and uniform conditional on E_1. All probabilities in this proof are conditional on E_1 alone, not on the existence of a bad separation.

Let $W = \bigcup_{C\in\mathcal{C}_0} V(C)$. Since fewer than N_0 cycles were examined, $|W| < \frac{h}{16}$, so $\kappa(D'-W) \geq h - |W| > \frac{15h}{16}$. Every remaining cycle is contained in A_L or A_R. Let V_L and V_R be the unions of the vertex sets of these two families, respectively, so $V(D'-W) = V_L \dot\cup V_R$. Both sets are determined by E_1.

For any agreeing separation, a colour-j path P from a cycle in V_L is represented in $L'\cup X$. Indeed, $x_P \in A_L$, and the cycle containing P is disjoint from I_L, so there is a vertex $y_{P'} \in I_L$ with $y_{P'}x_P \in E(G)$, where $v_{P'} \in L'$. This gives the arc $v_{P'}v_P$ in D_j, so $v_P \notin R'$, and hence $v_P \in L'\cup X$. Similarly, if P comes from a cycle in V_R, then there is a vertex $x_{P'} \in I_R$ with $y_Px_{P'} \in E(G)$ and $v_{P'} \in R'$. Thus $v_Pv_{P'} \in A(D_j)$, so $v_P \notin L'$, and hence $v_P \in R'\cup X$.

First suppose that $|V_L| < \frac{h}{4}$. For every $v \in V_L \cup W$,

$$|N^+_{D'}(v) \cap V_R| \geq d^+_U(v) - |V_L| - |W| > \frac{11h}{16}.$$

The number of colour-j paths in V_R whose first vertices are outneighbours of v is a sum of independent Bernoulli variables, with mean $\mu_v > \frac{11h}{16t\ell} = 11\cdot 2^{12}ta\ell \geq 2a$. Its probability of being less than a is at most $e^{-\frac{\mu_v}{8}} \leq e^{-\frac{h}{16t\ell}}$. Since $|V_L\cup W| < h$, except with probability at most $he^{-\frac{h}{16t\ell}}$, every vertex of $V_L\cup W$ sends arcs to the first vertices of at least a colour-j paths in V_R. For any agreeing bad separation, the last vertex of a path represented in L' lies in $V_L\cup W$. One of its at least a target paths in V_R avoids X, and hence is represented in R', a contradiction.

If $|V_R| < \frac{h}{4}$, apply the same argument to inneighbours. Except with probability at most $he^{-\frac{h}{16t\ell}}$, the first vertex of every path represented in R' receives arcs from the last vertices of at least a colour-j paths in V_L. At least one of those paths is represented in L', again a contradiction.

Suppose next that $|V_L|, |V_R| \geq \frac{h}{4}$. Since $\frac{h}{8} = 2^{13}t^2a\ell^2$ is an integer and $D'-W$ is $(\frac{h}{8})$-strongly connected, Lemma 2.6 gives a matching of $\frac{h}{8}$ arcs from V_L to V_R. Choose this matching using a fixed ordering, so that it depends only on E_1. Partition V_L and V_R into their original cycles. On each cycle in V_L, select the last vertex of its path if its colour is j, and no vertex otherwise; on each cycle in V_R, use the first vertex instead. The selections on different cycles are independent, and each vertex is selected with probability $\frac{1}{t\ell}$. Lemma 2.7 therefore applies to the matching. If Y is the number of surviving arcs, it gives

$$\mu = \mathbb{E}[Y \mid E_1] = \frac{h}{8t^2\ell^2} = 2^{13}a \geq 2a, \qquad \mathbb{P}[Y < a \mid E_1] \leq e^{-\frac{h}{64t^2\ell^2}}.$$

The surviving arcs use different cycles on each side, so they give vertex-disjoint arcs of D_j. If $Y \geq a$, the set X cannot meet all these vertex-disjoint arcs, because $|X| < a$. An arc avoiding X goes from L' to R', a contradiction. Combining the small-side and large-side bounds proves the claim. □

For a fixed word, different completed procedures that stop before N_0 examinations without discarding the word correspond to disjoint events. Claim 5 gives the same conditional upper

bound for each such event, so the total contribution from all procedures stopping before N_0 is at most $2he^{-\frac{h}{16t\ell}} + e^{-\frac{h}{64t^2\ell^2}}$. The only remaining possibility is that N_0 cycles are examined without discarding the word, whose probability is bounded by Claim 4. Therefore, for each fixed word, the probability that the word is not discarded and some bad separation agrees with the procedure is at most

$$e^{-\frac{h}{256t^2\ell}} + 2he^{-\frac{h}{16t\ell}} + e^{-\frac{h}{64t^2\ell^2}} \le 3e^{-\frac{h}{512t^2\ell^2}}. \tag{3.4}$$

We verify the last inequality. Since $\log(2h) = 17\log 2 + 2\log t + \log a + 2\log \ell \le 32ta\ell \le \frac{h}{32t\ell}$, we have $2he^{-\frac{h}{16t\ell}} \le e^{-\frac{h}{32t\ell}}$. Now let $z = \frac{h}{512t^2\ell^2} = 128a$. The three terms on the left of (3.4) are therefore at most $e^{-2\ell z}, e^{-16t\ell z}, e^{-8z}$, respectively. Each is at most e^{-z}, which proves (3.4).

Finally, Claim 3 gives 3^{2a-1} possible words for each colour. Since $3\ell \le 6a \le e^{2a}$ and $3^{2a-1} < e^{3a}$, a union bound over all words and colours gives

$$\begin{aligned}&\mathbb{P}[E_0 \text{ and some } D_j \text{ is not } a\text{-strongly connected}]\\ &\quad\le 3\ell\, 3^{2a-1}e^{-\frac{h}{512t^2\ell^2}} = 3\ell\, 3^{2a-1}e^{-128a} < e^{-123a} < \frac{1}{16}.\end{aligned}$$

Combining this with (3.2) gives probability greater than $\frac{7}{8}$ that all required properties hold, since $d_0 > a$. □

4 Proofs of the main results

We use the directed version of Menger's theorem.

Theorem 4.1 (Menger's theorem; see [3, 6])**.** *Let D be a finite digraph, and let $X, Y \subseteq V(D)$ be disjoint. Then the maximum number of pairwise vertex-disjoint directed X–Y paths equals the minimum size of a vertex set meeting every directed X–Y path.*

Lemma 4.2. *Let D be a digraph, let $d \ge 1$ be an integer, and let D_0 be a d-strongly connected digraph whose vertices v_P represent pairwise vertex-disjoint directed paths P in D. Assume that $y_P x_{P'} \in A(D)$ whenever $v_P v_{P'} \in A(D_0)$. Let $X, Y \subseteq V(D)$ be disjoint sets with $|X| = |Y| = r, 1 \le r \le d$, and suppose that $X \cup Y$ is disjoint from every represented path. Assume further that every $x \in X$ sends an arc to the first vertex of at least d represented paths, and every $y \in Y$ receives an arc from the last vertex of at least d represented paths.*

Then D contains r pairwise vertex-disjoint directed X–Y paths such that, after deleting their two endpoints, each of these paths is obtained by concatenating one or more represented paths, each used in its entirety.

Proof. Form an auxiliary digraph D_0' from D_0 by adding the vertices of X as sources and the vertices of Y as sinks. For $x \in X$ and $v_P \in V(D_0)$, add the arc xv_P whenever $xx_P \in A(D)$, and for $y \in Y$ add the arc $v_P y$ whenever $y_P y \in A(D)$. No other arcs incident with $X \cup Y$ are added.

Let $S \subseteq V(D_0')$ satisfy $|S| < r$, and let $S_0 = S \cap V(D_0)$. Since $|X| = |Y| = r > |S|$, there are vertices $x \in X \setminus S$ and $y \in Y \setminus S$. By assumption, x has at least d outneighbours in $V(D_0)$ and y has at least d inneighbours in $V(D_0)$. After deleting S_0, each therefore still has at least $d - |S_0| > 0$ such neighbors.

Moreover, $|S_0| \le |S| < r \le d$. Since D_0 is d-strongly connected, $D_0 - S_0$ is strongly connected. Choose an outneighbour v_P of x and an inneighbour v_Q of y in $D_0 - S_0$. There is a directed v_P–v_Q path in $D_0 - S_0$. Adding the arcs from x and to y gives a directed x–y path in $D_0' - S$.

Thus no set of fewer than r vertices meets every directed X–Y path in D_0'. By Theorem 4.1, D_0' contains r pairwise vertex-disjoint directed X–Y paths. Because no direct arc from X to Y was added, each such path contains at least one vertex of D_0. Since vertices of X are sources and vertices of Y are sinks, its only vertices in $X \cup Y$ are its initial and final vertices.

Finally, replace every auxiliary vertex v_P occurring on these paths by the whole represented path P, and replace each auxiliary arc $v_P v_{P'}$ by the corresponding joining arc $y_P x_{P'}$ of D. This produces r directed X–Y paths in D. They remain pairwise vertex-disjoint because the represented paths are pairwise vertex-disjoint and are disjoint from $X \cup Y$. □

Proof of Theorem 1.2. Let $\ell = \lceil \log_2 a \rceil + 1$ and $h = 2^{18} k^2 a \ell^2$, so $2k \geq 4$, $2k \not\equiv 0 \pmod 3$ and $\ell \leq 2a$. Let D satisfy $\alpha_2(D) \leq a$ and $\kappa(D) \geq r_k(a) + h$, and let $G = G_2(D)$. Take an inclusion-maximal family $\mathcal{C}$ of vertex-disjoint copies of C_{2k} in G, and let R be the set of uncovered vertices. Maximality implies that $G[R]$ contains no C_{2k}, so Lemma 2.3 gives $|R| \leq r_k(a)$. Since D is $(r_k(a) + h)$-strongly connected, deleting R leaves a digraph that is at least $r_k(a) + h - |R| \geq h$ strongly connected. Hence $D' = D - R$ is h-strongly connected. Let $U = V(D')$.

Moreover, every vertex $v \in V(D)$ satisfies $d_D^+(v), d_D^-(v) \geq r_k(a) + h$. Deleting the vertices of R removes at most $|R|$ inneighbours or outneighbours of v. Therefore, for every $v \in V(D)$,

$$d_U^+(v), d_U^-(v) \geq r_k(a) + h - |R| \geq h.$$

In particular, D' and $\mathcal{C}$ are nonempty.

Before making any random choices, apply Lemma 2.4 in $G[R]$ to the edgeless spanning linear forest on R, obtaining a path cover $\mathcal{R}$ with at most a paths. Let T be the set of endpoints of the paths in $\mathcal{R}$, so $T \subseteq R$ and $|T| \leq 2a$. By Lemma 3.1, choose retained edges, oriented complementary paths and colours with the stated connectivity and neighbour properties. Let $M = \{e_C : C \in \mathcal{C}\}$, let $V(M)$ be the set of endpoints of these edges, and let $Z = R \cup V(M)$.

Claim 6. *There is a path cover $\{P_1, \dots, P_p\}$ of $D[Z]$, where $1 \leq p \leq a$, containing one orientation of every edge of M. Every endpoint of these paths belongs to $T \cup V(M)$. In particular, the paths are disjoint from all complementary paths.*

Proof. The paths of $\mathcal{R}$ together with the edges of M form a spanning linear forest of $G[Z]$. Apply Lemma 2.4 to extend this forest to one with at most $\alpha(G[Z]) \leq a$ components, without creating new endpoints or deleting any edge. Orient each component in either direction, which is possible because all its edges belong to $G_2(D)$. Its endpoints are old endpoints, lying in $T \cup V(M)$, and all complementary paths avoid Z. □

We call the chosen orientations of the retained edges in M the *prescribed arcs*. Figure 1 gives a schematic illustration of the construction: the initial paths $P_1, \dots, P_p$ are successively joined into a directed cycle, which is then extended to a Hamilton cycle.

Claim 7. *Let $P_1, \dots, P_q$ be pairwise vertex-disjoint directed paths, where $2 \leq q \leq a$, such that $Z \subseteq \bigcup_{i=1}^{q} V(P_i)$ and every prescribed arc belongs to one of these paths. Suppose that all their endpoints lie in $T \cup V(M)$, and that their vertices outside Z form a union of complementary paths of previously used colours, each traversed in its entirety and in its selected direction.*

For any unused colour j, these paths can be joined into $\lceil \frac{q}{2} \rceil$ pairwise vertex-disjoint directed paths satisfying the same containment and traversal conditions, with colour j added to the used colours. Every endpoint of a resulting path is an endpoint of one of the original paths.

Proof. Let $r = \lfloor \frac{q}{2} \rfloor$. Choose $2r$ of the current paths and divide them into two families of r paths each; if q is odd, leave the remaining path unchanged. Let X be the set of last vertices of the paths in the first family, and let Y be the set of first vertices of the paths in the second family. Since the current paths are pairwise vertex-disjoint, X and Y are disjoint and $|X| = |Y| = r$.

Fix an unused colour j. By assumption, the current paths use only vertices of Z and complementary paths of previously used colours. Hence every path in $\mathcal{P}_j$ is disjoint from all current paths. Moreover, all vertices of $X \cup Y$ lie in $T \cup V(M)$, so Lemma 3.1 gives

$$d_j^+(x) \geq a \quad \text{for all } x \in X, \qquad d_j^-(y) \geq a \quad \text{for all } y \in Y.$$

Since $r \le q \le a$ and D_j is a-strongly connected, we may apply Lemma 4.2 with $d = a$. It gives r pairwise vertex-disjoint directed X–Y paths whose internal vertices are obtained from whole complementary paths of colour j.

Use these paths to join each path in the first family to a distinct path in the second family. Each such concatenation is a directed path, since the two old paths are distinct and the interior of the connecting path is disjoint from every current path. The resulting paths are pairwise vertex-disjoint, and they are also disjoint from the unchanged path when q is odd.

No old vertex or arc is removed, so the new paths still cover Z and contain every prescribed arc. Every newly added complementary path is used in its entirety and in its selected direction, and the endpoints of each new path are endpoints of the old paths. Thus all required properties are preserved. Finally, the number of resulting paths is $r + (q - 2r) = \lceil \frac{q}{2} \rceil$. □

Starting with Claim 6, apply Claim 7 while more than one path remains, using colours $1, 2, \dots$ in order. After i rounds there are at most $\lceil \frac{p}{2^i} \rceil$ paths, so at most $\lceil \log_2 a \rceil = \ell - 1$ rounds leave a single directed path. Its endpoints still lie in $T \cup V(M)$ and are distinct, since it contains a prescribed arc. Using the unused colour ℓ, apply Lemma 4.2 with $d = a$ and $r = 1$ to join its last vertex to its first. The connecting path is internally disjoint from the current path, so their union is a simple directed cycle C^* containing Z and every prescribed arc. Every complementary path is used by C^* either in its entirety or not at all.

Claim 8. *The cycle C^* extends to a Hamilton cycle without changing the cyclic order of its existing vertices.*

Proof. For each unused complementary path P_C, let uv be the orientation of its retained edge e_C on C^*. Replace the arc uv by the other u–v path along the original cycle C, directed from u to v. All edges of C are available in both directions in D, so this replacement is valid even if it traverses P_C opposite to its previously selected direction. The inserted internal vertices are exactly $V(P_C)$, all absent from C^*, and replacements for different cycles have disjoint interiors because the cycles of $\mathcal{C}$ are vertex-disjoint. All vertices of R already lie on C^*, and every remaining vertex belongs to one of these inserted paths, so the resulting directed cycle is Hamiltonian. □

This proves Theorem 1.2. □

Proof of Theorem 1.1. Let $t = \log(a + 2)$ and first assume that a is sufficiently large. Choose the least integer k with $s = k - 1 \ge \frac{t}{2 \log t}$ and $k \not\equiv 0 \pmod 3$. At most one integer is skipped because of divisibility by 3, so $s = \frac{t}{2 \log t} + O(1)$. For large a we have $\frac{\log a}{s} \le 2 \log t$ and $\frac{\log(2s)}{s} \le 1$, and hence

$$(2sa)^{1+\frac{1}{s}} = 2sa \exp\left(\frac{\log a + \log(2s)}{s}\right) \le 2e\, sat^2 = O\left(\frac{at^3}{\log t}\right).$$

The remaining summands in $r_k(a)$ are $O(ka)$, and the ceiling adds less than one, so $r_k(a) = O(\frac{at^3}{\log t})$. Also $2k = O(\frac{t}{\log t})$ and $\ell = O(t)$, giving

$$h_k(a) = 2^{18} k^2 a \ell^2 = O\left(\frac{at^4}{(\log t)^2}\right).$$

Since $\frac{at^3}{\log t} = o(\frac{at^4}{(\log t)^2})$, Theorem 1.2, together with $t \sim \log a$ and $\log t \sim \log \log a$, proves the claimed bound for all sufficiently large a. For the finitely many remaining integers $a \ge 3$, use Theorem 1.2 with $k = 2$ and enlarge the absolute constant c. □

Proof of Corollary 1.3. For each fixed $\varepsilon > 0$, $\frac{(\log a)^4}{(\log \log a)^2} = o(a^\varepsilon)$ as $a \to \infty$. The assertion follows from Theorem 1.1. □

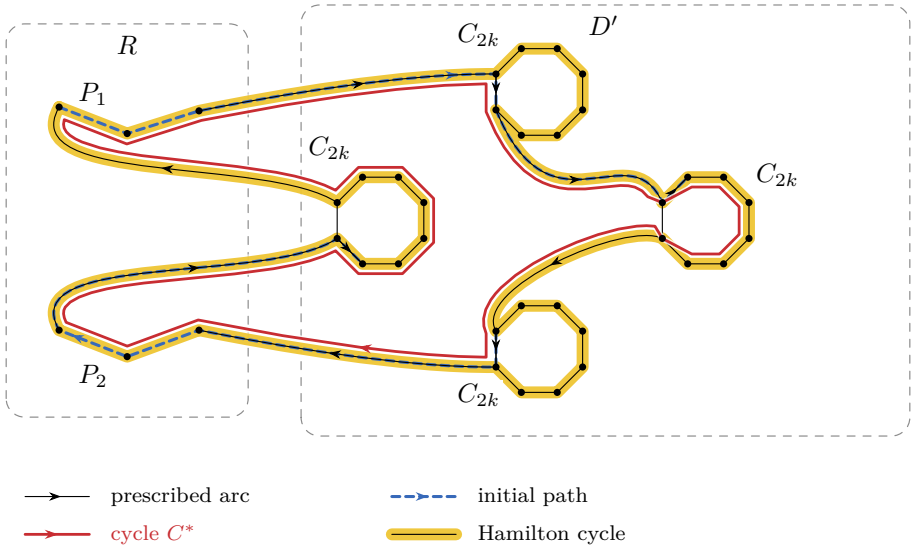


Figure 1: A schematic illustration of the construction of the Hamilton cycle in the proof of Theorem 1.2.

5 A counterexample to the Jackson–Ordaz pancyclicity conjecture for digraphs

We use the graph construction of Kostochka, Pudlák and Rödl [24, Section 2], in the affine-plane form described by Kocbek [23, Definition 2.2 and Theorem 4.2]. We describe the construction through a bipartite graph and prove its required properties before constructing the following digraphs.

Theorem 5.1. *For every prime $q \geq 11$, there is a Hamiltonian digraph D_q with no directed 3-cycle such that*

$$|V(D_q)| = 4q^2(q+1), \qquad \kappa(D_q) = q^2(q+1), \qquad \alpha_2(D_q) \leq 8q^2 + 4q - 4.$$

Let q be a prime, and let $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$ denote the field of integers modulo q. Define a bipartite graph G_q with vertex classes

$$U = \{u_{x,y} : x, y \in \mathbb{F}_q\}, \quad W = \{w_{a,b} : a \in \mathbb{F}_q \cup \{\infty\},\ b \in \mathbb{F}_q\}.$$

Here ∞ is an additional label, not a field element. For $a, b \in \mathbb{F}_q$, join $u_{x,y}$ to $w_{a,b}$ if and only if $y = ax + b$, and join $u_{x,y}$ to $w_{\infty,b}$ if and only if $x = b$. There are no other edges. Thus G_q is the point–line incidence graph of the affine plane over $\mathbb{F}_q$, with U representing points and W representing lines.

Use $0, 1, \ldots, q-1$ as the representatives of the field elements. Order U lexicographically, writing $u_{x,y} \prec u_{x',y'}$ if $x < x'$, or if $x = x'$ and $y < y'$, where the comparisons are between these integer representatives. Define a graph H_q with one vertex z_{uw} for each edge $uw \in E(G_q)$, where $u \in U$ and $w \in W$. For $uw, u'w' \in E(G_q)$ with $u \prec u'$, define adjacency by

$$z_{uw}z_{u'w'} \in E(H_q) \quad \Longleftrightarrow \quad w \neq w' \text{ and } uw' \in E(G_q). \tag{5.1}$$

Vertices corresponding to edges with the same endpoint in U are nonadjacent.

We first record the properties of G_q needed below.

Lemma 5.2. *The graph G_q has $|U| = q^2$ and $|W| = q(q+1)$. Every vertex in U has degree $q+1$, and every vertex in W has degree q. Any two distinct vertices in U have exactly one common neighbour. Consequently, G_q contains no 4-cycle and $|V(H_q)| = |E(G_q)| = q^2(q+1)$.*

Proof. The sizes of U and W follow from their definitions. For $u_{x,y} \in U$, we have

$$N_{G_q}(u_{x,y}) = \{w_{a,y-ax} : a \in \mathbb{F}_q\} \cup \{w_{\infty,x}\}.$$

These $q+1$ neighbours are distinct. For $a, b \in \mathbb{F}_q$, the neighbours of $w_{a,b}$ are precisely the q vertices $u_{x,ax+b}$ with $x \in \mathbb{F}_q$. The neighbours of $w_{\infty,b}$ are precisely the q vertices $u_{b,y}$ with $y \in \mathbb{F}_q$. This proves the degree assertions.

Consider distinct vertices $u_{x,y}, u_{x',y'} \in U$. If $x = x'$, then $y \neq y'$, and $w_{\infty,x}$ is a common neighbour. No $w_{a,b}$ with $a \in \mathbb{F}_q$ can be adjacent to both, since this would give $y = ax + b = y'$. Thus $w_{\infty,x}$ is their unique common neighbour. If $x \neq x'$, there is no common neighbour of the form $w_{\infty,b}$. The equations $y = ax + b$ and $y' = ax' + b$ have the unique solution $a = (y - y')(x - x')^{-1}$ and $b = y - ax$, giving exactly one common neighbour $w_{a,b}$. A 4-cycle would give two vertices in U with two distinct common neighbours, so G_q contains no 4-cycle.

Finally, summing the degrees over U gives $|E(G_q)| = q^2(q+1)$, and $|V(H_q)| = |E(G_q)|$ by construction. □

The construction of H_q is a special case of the ordered superline-graph construction of Kostochka, Pudlák and Rödl [24]; its affine-plane form was described by Kocbek [23]. The triangle-free property and the independence-number bound below also follow from the general construction in [24]. For completeness, we give direct proofs.

Lemma 5.3. *The graph H_q is triangle-free.*

Proof. Suppose that $z_{u_1w_1}, z_{u_2w_2}, z_{u_3w_3}$ form a triangle in H_q. By the definition of adjacency in H_q, the vertices u_1, u_2, u_3 are pairwise distinct, and the vertices w_1, w_2, w_3 are also pairwise distinct. Relabel them so that $u_1 \prec u_2 \prec u_3$. Their pairwise adjacencies and (5.1) give $u_1w_2, u_1w_3, u_2w_3 \in E(G_q)$. Also, $u_2w_2 \in E(G_q)$ by the definition of $z_{u_2w_2}$. Thus $u_1w_2u_2w_3u_1$ is a 4-cycle in G_q, contrary to Lemma 5.2. □

Lemma 5.4. *Every independent set in H_q corresponds to a set of edges forming a forest in G_q. In particular, $\alpha(H_q) \leq 2q^2 + q - 1$.*

Proof. Let $I \subseteq V(H_q)$ be independent, and let G_I be the spanning subgraph of G_q with edge set $\{uw \in E(G_q) : z_{uw} \in I\}$. Suppose that G_I contains a cycle C. Choose the smallest vertex u of $U \cap V(C)$ in the order $\prec$, and let w, w' be its two neighbours on C. Let u' be the neighbour of w' on C other than u. Then $u \prec u'$ and $w \neq w'$. Since $uw, uw', u'w' \in E(G_I)$, the vertices $z_{uw}, z_{u'w'} \in I$ are adjacent by (5.1), a contradiction. Hence G_I is a forest, and

$$|I| = |E(G_I)| \leq |V(G_q)| - 1 = 2q^2 + q - 1.$$ □

We now use four copies of H_q to construct the required digraph.

Proof of Theorem 5.1. Let $q \geq 11$ be a prime and let $m = q^2(q+1) = |V(H_q)|$. Take four vertex-disjoint copies of H_q, with vertex sets V_0, V_1, V_2, V_3. Replace every edge within each copy by two opposite arcs. For each $i \in \{0, 1, 2, 3\}$, add all arcs from V_i to V_{i+1}, where the subscripts of the parts are read modulo 4. Add no other arcs between distinct parts, and denote the resulting digraph by D_q. Then $|V(D_q)| = 4m$.

No directed 2-cycle meets two distinct parts, so $G_2(D_q)$ is the disjoint union of four copies of H_q. By Lemma 5.4, we have $\alpha_2(D_q) = 4\alpha(H_q) \leq 8q^2 + 4q - 4$.

We next prove that $\kappa(D_q) = m$. Let $S \subseteq V(D_q)$ with $|S| < m$. Every $V_i \setminus S$ is nonempty. Take distinct vertices $x, y \in V(D_q) \setminus S$, with $x \in V_i$, and choose $\eta \in \{1, 2, 3, 4\}$ so that $y \in V_{i+\eta}$. Let $x_0 = x$ and $x_\eta = y$, and choose $x_j \in V_{i+j} \setminus S$ for $1 \leq j < \eta$. These vertices are distinct, and $x_{j-1}x_j \in A(D_q - S)$ for every $1 \leq j \leq \eta$. Hence $x_0x_1 \cdots x_\eta$ is a directed x–y path in $D_q - S$. Thus $D_q - S$ is strongly connected and $\kappa(D_q) \geq m$. Conversely, in $D_q - V_0$ no arc leaves V_3 for another part, so no vertex of V_3 can reach V_1. Therefore $\kappa(D_q) \leq |V_0| = m$.

Any directed cycle meeting distinct parts must visit all four parts, since arcs between parts go only from V_i to V_{i+1}. Hence a directed 3-cycle would lie in a single part, and its vertices would form a triangle in H_q, contrary to Lemma 5.3.

Finally, write $V_i = \{v_{i,1}, \dots, v_{i,m}\}$. The m vertex-disjoint directed paths $v_{0,j}v_{1,j}v_{2,j}v_{3,j}$, for $1 \le j \le m$, together with the arcs $v_{3,j}v_{0,j+1}$ for $1 \le j < m$ and $v_{3,m}v_{0,1}$, form a Hamilton cycle in D_q. □

Proof of Theorem 1.4. For the digraphs in Theorem 5.1,

$$\frac{\kappa(D_q)}{\alpha_2(D_q)} \ge \frac{q^2(q+1)}{8q^2+4q-4} = \frac{q^2}{8q-4} > \frac{q}{8}.$$

Given $c > 0$, choose a prime $q \ge 11$ with $q > 8c$. Then D_q is Hamiltonian and satisfies $\kappa(D_q) > c\alpha_2(D_q)$, but it has no directed 3-cycle and is therefore not pancyclic. Moreover, $q \ge 11$ gives $\kappa(D_q) \ge \alpha_2(D_q) + 1$, so these digraphs also disprove the pancyclicity conjecture of Jackson and Ordaz for digraphs. □

References

[1] J. Ai and Y. Shi, A cubic refinement of Jackson's Chvátal–Erdős condition for Hamilton cycles in digraphs, preprint (2026), arXiv:2606.08401.

[2] N. Alon and J. H. Spencer, *The Probabilistic Method*, fourth ed., Wiley, Hoboken, NJ, 2016.

[3] J. Bang-Jensen and G. Gutin, *Digraphs: Theory, Algorithms and Applications*, second ed., Springer, London, 2009.

[4] J. A. Bondy, Pancyclic graphs I, *J. Combin. Theory Ser. B* **11** (1971), 80–84.

[5] J. A. Bondy, Pancyclic graphs: recent results, in *Infinite and Finite Sets*, Vol. I (Colloq., Keszthely, 1973), Colloq. Math. Soc. János Bolyai **10**, North-Holland, Amsterdam, 1975, 181–187.

[6] J. A. Bondy and U. S. R. Murty, *Graph Theory*, Graduate Texts in Mathematics **244**, Springer, London, 2008.

[7] Y. Caro, New results on the independence number, Technical Report, Tel-Aviv University, 1979.

[8] G. Chen, R. J. Faudree and R. J. Gould, Intersections of longest cycles in k-connected graphs, *J. Combin. Theory Ser. B* **72** (1998), 143–149.

[9] V. Chvátal and P. Erdős, A note on Hamiltonian circuits, *Discrete Math.* **2** (1972), 111–113.

[10] N. Draganić, D. Munhá Correia and B. Sudakov, Chvátal–Erdős condition for pancyclicity, *J. Assoc. Math. Res.* **2** (2024), 1–14.

[11] N. Draganić, D. Munhá Correia and B. Sudakov, Pancyclicity of Hamiltonian graphs, *J. Eur. Math. Soc.* **28** (2026), 4047–4062.

[12] D. Dubhashi and D. Ranjan, Balls and bins: a study in negative dependence, *Random Structures Algorithms* **13** (1998), 99–124.

[13] P. Erdős, Some problems in graph theory, in *Hypergraph Seminar* (Ohio State Univ., Columbus, 1972), Lecture Notes in Mathematics **411**, Springer, Berlin, 1974, 187–190.

[14] T. Gallai and A. N. Milgram, Verallgemeinerung eines graphentheoretischen Satzes von Rédei, *Acta Sci. Math. (Szeged)* **21** (1960), 181–186.

[15] A. Ghouila-Houri, Une condition suffisante d'existence d'un circuit hamiltonien, *C. R. Acad. Sci. Paris* **251** (1960), 495–497.

[16] R. Häggkvist and C. Thomassen, On pancyclic digraphs, *J. Combin. Theory Ser. B* **20** (1976), 20–40.

[17] J. Han, On vertex-disjoint paths in regular graphs, *Electron. J. Combin.* **25** (2018), Paper No. 2.12, 6 pp.

[18] B. Jackson, A Chvátal–Erdős condition for Hamilton cycles in digraphs, *J. Combin. Theory Ser. B* **43** (1987), 245–252.

[19] B. Jackson and O. Ordaz, Chvátal–Erdős conditions for paths and cycles in graphs and digraphs: a survey, *Discrete Math.* **84** (1990), 241–254.

[20] K. Joag-Dev and F. Proschan, Negative association of random variables with applications, *Ann. Statist.* **11** (1983), 286–295.

[21] P. Keevash, D. Kühn and D. Osthus, An exact minimum degree condition for Hamilton cycles in oriented graphs, *J. London Math. Soc.* (2) **79** (2009), 144–166.

[22] P. Keevash and B. Sudakov, Pancyclicity of Hamiltonian and highly connected graphs, *J. Combin. Theory Ser. B* **100** (2010), 456–467.

[23] M. Kocbek, Explicit geometric construction of Ramsey graphs, preprint (2025), arXiv:2507.09235.

[24] A. Kostochka, P. Pudlák and V. Rödl, Some constructive bounds on Ramsey numbers, *J. Combin. Theory Ser. B* **100** (2010), 439–445.

[25] D. Kühn and D. Osthus, A survey on Hamilton cycles in directed graphs, *European J. Combin.* **33** (2012), 750–766.

[26] S. Letzter, Pancyclicity of highly connected graphs, *J. Eur. Math. Soc.*, to appear, DOI 10.4171/JEMS/1808; arXiv:2306.12579.

[27] A. Lo, V. Patel and M. A. Yıldız, Cycle partitions in dense regular digraphs and oriented graphs, *Forum Math. Sigma* **13** (2025), e79, 31 pp.

[28] O. Pikhurko, A note on the Turán function of even cycles, *Proc. Amer. Math. Soc.* **140** (2012), 3687–3692.

[29] V. K. Wei, A lower bound on the stability number of a simple graph, Bell Laboratories Technical Memorandum 81-11217-9, 1981.